

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – JIGSAW ACTIVITY

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO4 – Articulation (BL3)	Assessment:	Assessment Tool 2 – Jigsaw Activity
Weightage:	35% of CIA	Total Marks:	100
CO4 Statement:	<i>Apply hashing strategies for efficient data storage and sorting algorithms for arrangement of data.</i>		
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Section / Batch:	B Tech AIDS	Date:	19/08/2024

Code of Conduct

I _____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: _____

Instructions

- Home Groups (8): 5 students, Expert Groups: One expert per topic. Total: 100 Marks.
- Students first discuss in Expert Groups. They return to Home Groups and teach teammates.
- Each student teaches Concept, Working, Advantages, Disadvantages, Applications, Time Complexity.

Section / Topic	Focus	Q Nos	Marks
Hashing & Sorting	Hashing, Resolution Techniques, Sorting	8 Teams	100
GRAND TOTAL:		100 Marks	

Annexure A – Jigsaw Activity

1. Scenario 1 – Airport Baggage System (Every luggage has Barcode)
 - a. Hashing Method
 - b. Collision handling
 - c. Fast luggage
2. Scenario 2 – E-Commerce Warehouse (Products stored by SKU)
 - a. Hashing strategy
 - b. Inventory retrieval
 - c. Sorting inventory.
3. Scenario 3 – Traffic Police Database (Vehicle Registration Numbers)
 - a. Fast searching
 - b. Duplicate handling
 - c. Sorting violations
4. Scenario 4 – Social Media Platform (Millions of users)
 - a. Hash Function
 - b. Collision handling – Separate Chaining
 - c. Fast searching
5. Scenario 19 – University Admission Portal (A university receives 75,000 applications. Applicants are searched using Application IDs, while merit lists are prepared based on entrance scores.)
6. Scenario 22 – Pharmacy Inventory Management (A pharmacy maintains thousands of medicine records. Pharmacists search medicines by code, while stock reports are arranged based on expiry dates.)
7. Scenario 24 – Weather Monitoring System (A weather station collects data from thousands of sensors every minute. Scientists retrieve records using sensor IDs and generate reports sorted by temperature and rainfall.)
8. Scenario 6 – Digital Music Streaming App (Songs, Artist, Playlist, Genre)
 - a. Fast searching
 - b. Hash Function
 - c. Fast Sorting

